

Dohyun Lee

PH.D. STUDENT IN ARTIFICIAL INTELLIGENCE

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About

My current research interest lies in how robots can build and use an understanding of the 3D world to manipulate objects and act in real environments. My research centers on two directions. First, while building PUBG Ally at KRAFTON, I saw first-hand that vision-language models understand complex 3D scenes remarkably well, and I now aim to bring this capability to robot manipulation through explicit 3D spatial reasoning, together with tools such as SAM 3 and GraspGen-X. Second, video is another rich source of 3D world understanding. I believe that video generation is essentially representation learning for the 3D world and its dynamics, and I am currently studying what learning objectives yield good representations for world action models.

I am currently a Ph.D. student at the DAVIAN Lab, KAIST AI, advised by Prof. Jaegul Choo. Previously, I was a research intern at KRAFTON.

Education

Korea Advanced Institute of Science and Technology

[Seongnam, S.Korea](#)

INTEGRATED M.S./PH.D. IN ARTIFICIAL INTELLIGENCE (ADVISOR: PROF. JAEGUL CHOO, DAVIAN LAB)

Sep. 2023 - Feb. 2029

(Expected)

Kyunghee University

[Yongin, S.Korea](#)

B.S. IN MECHANICAL ENGINEERING AND SOFTWARE ENGINEERING

Mar. 2017 - Aug. 2023

- GPA: 4.27/4.5 (Major: 4.29/4.5)
- 4 Academic Excellence Scholarship, 2 Academic Excellence Award

Research Experience

KRAFTON

[Seoul, S.Korea](#)

RESEARCH INTERN, AI COMPANION TEAM

Nov. 2024 - Feb. 2026

- Researched and developed PUBG Ally, showcased at CES 2025 in collaboration with NVIDIA ([LINK](#)).
- PUBG Ally was beta-launched in the live game PUBG: BATTLEGROUNDS ([LINK](#)).

Hyundai Motor Company

[Seoul, S.Korea](#)

UNDERGRADUATE NLP RESEARCHER

Jul. 2022 - Aug. 2022

Ongoing Projects

Soft-Value Flow: Exact Value Guidance for Flow-Matching Policies in Offline RL

DOHYUN LEE^{*}, JUNHYEONG HONG^{*}, THEO CHOI, KINAM KIM, JAEGUL CHOO, YOONSEON OH

2026

Submitted to International Conference on Learning Representations (ICLR), 2027

Vision-Language Models as Strong Pick-and-Place Manipulators with 3D Reasoning

DOHYUN LEE, KINAM KIM, JUNHA HYUNG, DONGYOON HWANG, JAEGUL CHOO

2026

Representation Learning for World Action Models

DOHYUN LEE, DONGYOON HWANG, MINHO PARK, KINAM KIM, TAEWOONG KANG, JUNHA HYUNG, JAEGUL CHOO

2026

Publications

Procedural Memory for Proactive Robot Assistance

D. LEE^{*}, M. PARK^{*}, J. PARK, D. HWANG, J. HYUNG, S. HWANG, B. LEE, J. CHOO

2026

Conference on Robot Learning (CoRL), 2026

PersonaTrail: Benchmarking Personalized Web Agents through Browsing Trails

S. YANG, C. KI, D. LEE, J. CHOO, C. PARK

2026

arXiv preprint, 2026

J. PARK, K. JUNG, **D. LEE**, H. LEE, D. GWAK, C. PARK, J. CHOO, J. CHO
Advances in Neural Information Processing Systems (NIPS), 2025

M. CHOI, D. RIM, **D. LEE**, J. CHOO 2025
Annual Meeting of the Association for Computational Linguistics, 2025

J. PARK, H. LEE, **D. LEE**, D. GWAK, J. CHOO
Annual Meeting of the Association for Computational Linguistics, 2025

D. LEE, S. LEE, C. YANG, Y. BAEK, J. CHOO
Findings of the Association for Computational Linguistics, 2025

M. CHOI, C. PARK, **D. LEE**, J. CHOO
arXiv preprint, 2024

C. PARK, M. CHOI, **D. LEE**, J. CHOO
Conference on Language Modeling, 2024

D. LEE*, D. RIM*, M. CHOI, J. CHOO 2024
Findings of the Association for Computational Linguistics, 2024